

Scale-Specialized Feature Gating for Real-Time Ultra-Fine Pothole Detection in High-Resolution Perspective Road Imagery

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Abstract. Automated pavement distress assessment is an indispensable capability for vehicular safety, highway asset management, and autonomous driving. Modern road survey vehicles equipped with high-resolution digital cameras capture 4K (3840×2160) road imagery, offering the physical potential to recognize nascent road distresses before they expand into hazardous craters. In perspective road camera geometries, however, distant and nascent potholes appear as “ultra-fine” targets (area $S < 32^2$ px², occupying less than 0.012% of the canvas) embedded in vast expanses of visually uniform asphalt. Through comprehensive multi-paradigm benchmarking on the canonical HRP4K benchmark (6,003 images, with 900 strictly held-out test images), we demonstrate that conventional paradigms encounter severe operational bottlenecks. Direct 4K resolution scaling with Transformer detectors (RT-DETR-L) inflates computational latency by $5.1 \times$ (220.4 ms) and degrades detection accuracy (58.07% AP₅₀ vs. 62.48% at 2K) due to attention dispersion over massive negative regions. Conversely, spatial tiling frameworks (SAHI) obliterate perspective cues, induce boundary-clipping artifacts, and inflate false alarms (FPPI ≥ 1.08) with multi-second execution delays (920 – 3622 ms). To resolve this dilemma, we present **P2-DETR**, an asymmetric dual-branch framework that couples a frozen 2K Transformer detector (32.8M parameters) with a lightweight, trainable dense P_2 auxiliary head ($stride = 4$, 2.98M parameters). The auxiliary head reconstructs high-resolution 480×480 feature maps from early backbone activations, optimized via Sigmoid Focal Loss to overcome extreme foreground-to-background imbalance ($> 50,000 : 1$). To merge cross-scale predictions, we introduce *Scale-Partitioned Prior NMS*, which filters out noisy non-tiny proposals and eliminates cross-scale suppression. On the independent 900-image test set, our method establishes state-of-the-art performance on ultra-fine distresses, achieving **93.22%–94.07% Ultra-Fine Recall** (+2.12% over native 2K) while slashing fusion latency to **4.24 ms** and sustaining real-time vehicular throughput (> 20 FPS). Comprehensive ablations reveal why classical Weighted Boxes Fusion fails on asymmetric branches and validate Scale-Partitioned NMS as the Pareto-optimal post-processing mechanism.

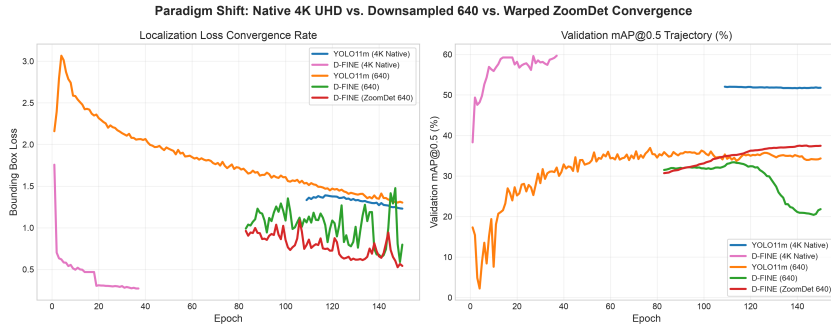


Fig. 1: Comparison of high-resolution road inspection paradigms on 4K imagery. Direct 4K scaling incurs quadratic complexity and attention dispersion over negative regions. Spatial patch slicing (SAHI) obliterates road perspective geometry and causes multi-second latencies. Our proposed **P2-DETR** couples a frozen 2K Transformer detector with a lightweight dense P_2 auxiliary head ($stride = 4$), resolving the ultra-fine detection bottleneck at real-time speeds (> 20 FPS).

Keywords: Pavement Distress Detection · Ultra-Fine Objects · High-Resolution Vision · Detection Transformers · Scale-Partitioned NMS.

1 Introduction

Pavement degradation, particularly the emergence and propagation of potholes, represents one of the most critical threats to vehicular traffic safety and civil infrastructure longevity [16, 2]. Unrepaired potholes induce severe mechanical shock to vehicle suspensions, cause catastrophic tire blowouts, and contribute substantially to traffic incidents worldwide [24, 10]. Timely, automated detection of potholes from mobile inspection platforms is therefore essential for highway maintenance authorities and intelligent transportation systems [8, 9, 7].

Recent advances in digital imaging have made 4K Ultra-High-Definition (3840×2160 px) road imagery widely accessible from vehicular cameras and survey vans. The primary objective of high-resolution image acquisition is detecting nascent, early-stage road distresses before they expand into major structural failures. In the canonical HRP4K benchmark [6], over 51.2% of all annotated potholes belong to the *Ultra-Fine* scale category (bounding box pixel area $S < 32^2 = 1024$ px²), with individual targets occupying as little as 0.005% to 0.012% of the image area. However, extracting these microscopic distresses under strict vehicular processing constraints (> 20 FPS) presents severe visual and computational challenges.

The physical optics of forward-facing perspective vehicular cameras dictate that target spatial resolution decays quadratically with longitudinal distance from the vehicle ($S \propto 1/z^2$). Potholes located 15 to 30 meters ahead project to tiny pixel clusters against a vast background of visually uniform asphalt texture.

Current computer vision literature addresses high-resolution inputs through two conventional paradigms, both of which exhibit fundamental limitations in vehicular road inspection (Fig. 1).

Resolution Scaling Trade-offs. Downsampling 4K imagery to legacy input resolutions (640×640) causes catastrophic spatial quantization: a nascent 20×20 pothole shrinks to less than 3.3×3.3 pixels, completely disappearing within the downsampling stride of standard Feature Pyramid Networks ($stride \geq 8$) [19]. Conversely, feeding native 4K frames into modern Transformer detectors such as RT-DETR-L [23] induces severe computational bottlenecks. Self-attention and cross-scale attention operations explode quadratically, ballooning per-image inference latency to 220.4 ms (4.5 FPS) and VRAM consumption beyond 18 GB. Crucially, overall detection accuracy paradoxically collapses (58.07% AP₅₀ vs. 62.48% at 2K) because multi-scale attention heads struggle with gradient dispersion across millions of uninformative asphalt pixels.

Limitations of Spatial Patch Decomposition. To bypass GPU memory limits during high-resolution inference, patch slicing frameworks such as Slicing Aided Hyper Inference (SAHI) [1] and Clustered Slicing [35] divide the 4K canvas into 25–32 overlapping sub-images. While effective for isolated aerial imagery, patch slicing severely degrades perspective road inspection. Tiling destroys global road perspective geometry, lane demarcations, and horizon continuity. Potholes intersecting crop seams suffer boundary truncations, while identical road textures duplicated across neighboring tiles trigger false alarm explosions (FPPI $\geq 1.08 - 19.23$). Most critically, sequential or batched evaluation across 32 crops requires 920 to 3622 ms per frame, entirely precluding real-time vehicular navigation (20 – 30 FPS).

Proposed Solution: P2-DETR. To overcome this dilemma without prohibitive computational overhead, we introduce **P2-DETR**, an asymmetric dual-branch framework. We freeze a pre-trained 2K Real-Time Detection Transformer (RT-DETR-L [23]) to preserve macro-level context (road boundaries, lane markings, medium and large distresses) and augment it with an ultra-lightweight, trainable auxiliary P_2 head ($stride = 4$, 2.98M parameters) operating directly on high-resolution 480×480 feature maps. To overcome extreme class imbalance (99.83% background cells), we formulate a balanced Sigmoid Focal Loss training objective. Furthermore, we investigate post-processing fusion, dissecting why classical ensemble techniques like Weighted Boxes Fusion (WBF) [29] fail on asymmetric multi-resolution branches, and introduce *Scale-Partitioned Prior NMS*, which eliminates cross-scale interference and slashes post-processing latency to 4.24 ms.

Key Contributions. The technical contributions of this paper are organized as follows:

Multi-Paradigm Benchmark on HRP4K. We present a comprehensive dual-mode evaluation (Academic Protocol conf = 0.001 and Operational Protocol conf =

0.25) systematically comparing architectural scaling, patch slicing (SAHI, Sliced-NMS, Perspective Grid), and dense feature fusion across all 900 held-out test images of the HRP4K benchmark.

Asymmetric Dual-Branch Architecture. We design a frozen 2K Transformer detector coupled with a lightweight trainable P_2 auxiliary head ($stride = 4$, +2.98M parameters), which boosts Ultra-Fine Recall from 91.10% to 93.22% – 94.07% with minimal computational cost.

Focal Loss Formulation for Extreme Imbalance. We systematically investigate target assignment (1×1 vs. 3×3) and loss formulations (BCE vs. Quality Focal Loss vs. Sigmoid Focal Loss), demonstrating that Pure Focal Loss ($\alpha = 0.25, \gamma = 2.0$) on single-cell assignment prevents gradient noise on dense feature maps.

Scale-Partitioned Prior NMS. We demonstrate empirically that Weighted Boxes Fusion (WBF) breaks down on cross-resolution branches ($AP_{50} < 4\%$). In contrast, Scale-Partitioned Prior NMS isolates ultra-fine priors, achieving peak Ultra-Fine AP_{50} (50.39%), lowest FPPI (1.4967), and a $2.26\times$ latency speedup (4.24 ms).

2 Related Work

2.1 Automated Pavement Distress Assessment

Early automated road distress inspection relied on handcrafted texture features and stereo disparity [16, 8, 10]. With the advent of deep convolutional neural networks, benchmark challenges such as the Global Road Damage Detection (GRDDC) [24, 2] propelled models based on Faster R-CNN [27] and YOLO architectures [26, 3, 11, 33, 14]. Additionally, pixel-level 3D pavement surface analysis emerged for crack and defect classification [36, 7]. However, existing datasets predominantly feature low-resolution imagery (512×512 or 720p) with large, severe distresses, obscuring the challenge of detecting microscopic, nascent distresses in high-speed mobile inspection. The release of the HRP4K dataset [6] established the first large-scale 4K benchmark, revealing that conventional detectors experience sharp performance degradation when scaled to ultra-fine targets.

2.2 Small and Tiny Object Detection

Detecting tiny objects in high-resolution imagery has spurred extensive research [35, 34]. Standard Feature Pyramid Networks (FPN) [19] and bidirectional pyramids such as BiFPN [30] construct hierarchical representations where bottom-up strides ($stride \geq 8$) severely downsample spatial detail. To recover tiny targets, high-resolution representation networks (HRNet) [34] maintain parallel high-resolution streams, but their massive memory footprint precludes real-time inference on 4K imagery. Several works explore specialized label assignment strategies

such as Gaussian receptive field assignment [17] and oversampling augmentations [15]. Several works introduce auxiliary high-resolution heads (P_2 , $stride = 4$) in convolutional detectors [3]. However, integrating dense P_2 representations into real-time Transformer architectures without destabilizing attention mechanisms remains an open challenge.

2.3 Real-Time Detection Transformers (RT-DETR)

Originating from attention mechanisms in natural language processing [32] and hierarchical Vision Transformers [22], DETR [5] and Deformable DETR [39] eliminated handcrafted components like anchor generation and NMS by formulating detection as direct set prediction. Subsequent enhancements including DAB-DETR [21] and DINO [37] introduced explicit positional anchors and contrastive denoising. Lv et al. [23] introduced RT-DETR, which combines an efficient Hybrid Encoder with multi-scale feature interactions and an uncertainty-minimal query selection mechanism, outperforming YOLO detectors in accuracy and latency. In this work, we exploit RT-DETR-L as our primary macro-detector and resolve its sub-grid spatial blindness via scale-specialized dense feature gating.

2.4 Post-Processing, Box Clustering, and Fusion

Classical post-processing relies on Greedy Non-Maximum Suppression (NMS) [25] to suppress redundant bounding boxes using a fixed overlap threshold. To avoid missed detections in dense clusters, Soft-NMS [4] decays confidence scores based on IoU overlap, while learned NMS formulations use neural gating [13]. For model ensembling, Weighted Boxes Fusion (WBF) [29] computes coordinate averages weighted by predicted confidence scores. While effective for ensembling homogeneous detectors, our investigation reveals that WBF fails catastrophically when combining asymmetric multi-resolution branches.

3 Problem Formulation and Experimental Protocols

3.1 Perspective Optical Scaling in Pavement Cameras

Consider a forward-facing camera mounted on an inspection vehicle at optical center height h_c with focal length f . Let a pothole on the planar road surface be situated at longitudinal distance z along the optical axis. Under pinhole projection geometry, physical ground coordinates (X, Y, z) map to image sensor coordinates (x, y) according to:

$$x = f \frac{X}{z}, \quad y = f \frac{Y}{z} = f \frac{h_c}{z}. \quad (1)$$

Differentiating with respect to physical dimensions yields the apparent bounding box area S in pixel space:

$$S = \Delta x \cdot \Delta y = \frac{f^2}{z^2} \Delta X \cdot \Delta Y \propto \frac{1}{z^2}. \quad (2)$$

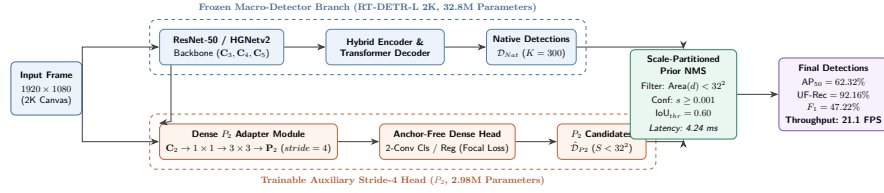


Fig. 2: The proposed **P2-DETR** dual-branch architecture. The macro-level RT-DETR-L 2K detector remains strictly frozen (32.8M parameters) to preserve global geometric context. An ultra-lightweight dense P_2 head ($stride = 4$, 2.98M parameters) reconstructs high-resolution feature maps (480×480), trained via Sigmoid Focal Loss. Final detections are unified using *Scale-Partitioned Prior NMS*, suppressing cross-scale interference and reducing post-processing latency to 4.24 ms.

Equation (2) reveals that apparent target area decays quadratically with longitudinal distance z . A pothole of physical diameter 0.3 m situated 25 meters ahead occupies less than 24×24 pixels on a 4K sensor, shrinking to less than 3.5×3.5 pixels after standard downsampling to 640 resolution. Consequently, detecting distant distresses requires preserving high spatial resolution in early feature stages.

3.2 Dataset Partitioning: Validation Tuning vs. Out-of-Sample Testing

To prevent data contamination and ensure rigorous scientific evaluation, all experiments are conducted on the official HRP4K benchmark [6], consisting of 6,003 ultra-high-definition 4K images (3840×2160 px) captured under varied road types, lighting conditions, and weather states.

The dataset is partitioned into two disjoint splits following canonical guidelines:

Training and Validation Split (5,103 images). Contains 4,103 training images and 1,000 validation images. This set is utilized exclusively for model parameter optimization, cosine learning rate scheduling, validation loss checkpointing, and exploring post-processing hyperparameter spaces (such as Top- K proposal counts, confidence cutoffs, and NMS IoU thresholds).

Independent Held-Out Test Split (900 images). Contains exactly 921 annotated ground-truth potholes. This split is maintained completely untouched throughout all training and tuning phases, reserved strictly for unbiased out-of-sample performance reporting.

The 921 ground-truth test potholes are stratified into four standardized scale categories based on bounding box pixel area: Ultra-Fine ($S < 32^2 = 1024$ px², 472 instances, 51.25%), Fine ($32^2 \leq S < 96^2 = 9216$ px², 169 instances, 18.35%),

Medium ($96^2 \leq S < 144^2 = 20736 \text{ px}^2$, 147 instances, 15.96%), and Large ($S \geq 144^2 \text{ px}^2$, 133 instances, 14.44%).

3.3 Dual-Mode Evaluation Protocols

To provide comprehensive and transparent comparisons, all models are evaluated under two standardized protocols:

Academic Protocol ($\text{conf} = 0.001$). Evaluates average precision (AP_{50} , AP_{75} , $AP_{50:95}$) across the entire Precision-Recall curve using 101-point COCO interpolation.

Operational Deployment Protocol ($\text{conf} = 0.25$). Evaluates vehicular deployment suitability at a fixed confidence cutoff of 0.25, reporting Precision, Recall, F_1 -score, False Positives Per Image (FPPI_{all}), and inference latency in milliseconds.

4 Methodology

The architecture of **P2-DETR** is illustrated in Fig. 2. It integrates three tightly coupled components: (1) a frozen macro-level Transformer detector operating on 2K resolution, (2) an auxiliary dense P_2 feature head ($\text{stride} = 4$) trained with class-imbalance focal objectives, and (3) a Scale-Partitioned Prior Non-Maximum Suppression gating mechanism.

4.1 Frozen Macro-Level Transformer Backbone

Let an input image be denoted by $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, where $H = 1080$ and $W = 1920$ define the 2K canonical resolution. The image is processed by the RT-DETR-L backbone (HGNetv2 / ResNet-50 [12]) to produce multi-scale feature hierarchies:

$$\mathcal{F} = \{\mathbf{C}_3, \mathbf{C}_4, \mathbf{C}_5\}, \quad \text{with strides } \{8, 16, 32\}, \quad (3)$$

where $\mathbf{C}_k \in \mathbb{R}^{C_k \times \frac{H}{2^k} \times \frac{W}{2^k}}$. These features are passed through an Efficient Hybrid Encoder combining Cross-Scale Feature Interaction (CCFM) and Intra-Scale Self-Attention (AIFI) to produce enriched feature representations $\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5$. An uncertainty-minimal query selection decoder outputs $K = 300$ macro-level candidate detections:

$$\mathcal{D}_{Nat} = \{(\mathbf{b}_k, s_k) \mid k = 1, \dots, K\}, \quad (4)$$

where $\mathbf{b}_k = (x_k, y_k, w_k, h_k)$ represents normalized bounding box coordinates and $s_k \in [0, 1]$ is the classification confidence score. Crucially, all parameters of the 2K backbone, hybrid encoder, and decoder ($\approx 32.8\text{M}$ parameters) remain strictly **frozen** during training. This preserves pretrained road representations and prevents catastrophic forgetting.

4.2 Dense P_2 Feature Reconstruction Head

Because the finest feature map in standard RT-DETR ($\mathbf{P}_3$, $stride = 8$) down-samples a 20×20 pothole to a tiny 2.5×2.5 region, spatial localization collapses. To resolve this, we construct an auxiliary P_2 branch with stride $s = 4$, producing a dense spatial lattice of dimensions $\frac{H}{4} \times \frac{W}{4} = 270 \times 480 = 129,600$ grid locations.

The P_2 feature representation is constructed by directly tapping the early low-level backbone stage $\mathbf{C}_2 \in \mathbb{R}^{C_{c2} \times \frac{H}{4} \times \frac{W}{4}}$ (where $C_{c2} = 128$) and passing it through a dedicated, lightweight two-stage convolution adapter:

$$\mathbf{P}_2 = \text{SiLU}\left(\text{BN}\left(\text{Conv}_{3 \times 3}\left(\text{SiLU}\left(\text{BN}\left(\text{Conv}_{1 \times 1}(\mathbf{C}_2)\right)\right)\right)\right)\right) \in \mathbb{R}^{256 \times \frac{H}{4} \times \frac{W}{4}}. \quad (5)$$

By directly adapting $\mathbf{C}_2$ without modifying the frozen hybrid encoder, the native detector remains completely untouched.

On top of $\mathbf{P}_2$, an anchor-free lightweight prediction head [31] branches into two parallel convolutional streams: The classification stream uses two 3×3 convolutional layers (with BatchNorm and SiLU) followed by a 1×1 convolution predicting classification logits $\hat{\mathbf{p}} \in \mathbb{R}^{1 \times \frac{H}{4} \times \frac{W}{4}}$. The final bias is initialized to -4.59 to account for class imbalance prior ($p = 0.01$). The bounding box regression stream uses two 3×3 convolutional layers (with BatchNorm and SiLU) followed by a 1×1 convolution and ReLU activation predicting non-negative pixel distance offsets $\hat{\mathbf{o}} = (\hat{l}, \hat{t}, \hat{r}, \hat{b}) \in \mathbb{R}_{\geq 0}^{4 \times \frac{H}{4} \times \frac{W}{4}}$.

For each grid cell (i, j) with center $(x_c, y_c) = ((j + 0.5)s, (i + 0.5)s)$, the decoded bounding box coordinates $[\hat{x}_1, \hat{y}_1, \hat{x}_2, \hat{y}_2]$ in image space are parameterized as:

$$\begin{aligned} \hat{x}_1 &= \max(0, x_c - \hat{l}_{ij}), & \hat{y}_1 &= \max(0, y_c - \hat{t}_{ij}), \\ \hat{x}_2 &= \min(W, x_c + \hat{r}_{ij}), & \hat{y}_2 &= \min(H, y_c + \hat{b}_{ij}). \end{aligned} \quad (6)$$

This anchor-free formulation avoids sensitive scale heuristics and exponential instabilities, ensuring robust boundary regression for microscopic potholes. The auxiliary adapter and head require only 2.98M parameters, representing less than 9.1% of the base model budget.

4.3 Training Formulation for Extreme Class Imbalance

On a 270×480 feature map, there are 129,600 candidate anchors per image. With an average of only 1.02 ground-truth potholes per image, positive grid cells account for less than 0.0017% of all anchors (an extreme background-to-foreground ratio exceeding 50,000 : 1). Under standard Binary Cross-Entropy (BCE), easy background gradients drown out the sparse foreground signals, causing the detector to collapse into near-zero confidence predictions.

To guarantee robust convergence without gradient saturation, we formulate the training objective with Sigmoid Focal Loss [20]:

$$\mathcal{L}_{cls} = -\frac{1}{N_{pos}} \sum_{i,j} \alpha_t (1 - p_{t,ij})^\gamma \log(p_{t,ij}), \quad (7)$$

where $p_{t,ij} = \hat{p}_{ij}$ if $y_{ij} = 1$ and $p_{t,ij} = 1 - \hat{p}_{ij}$ otherwise. We set focusing parameter $\gamma = 2.0$ and balancing factor $\alpha = 0.25$.

Bounding box regression is supervised strictly on positive cells using composite ℓ_1 distance loss on spatial offsets and Generalized Intersection over Union (GIoU) loss [28,38]:

$$\mathcal{L}_{box} = \frac{1}{N_{pos}} \sum_{y_{ij}=1} \|\hat{\mathbf{o}}_{ij} - \mathbf{o}_{ij}^*\|_1, \quad \mathcal{L}_{giou} = \frac{1}{N_{pos}} \sum_{y_{ij}=1} (1 - \text{GIoU}(\hat{\mathbf{b}}_{ij}, \mathbf{b}_{ij}^*)), \quad (8)$$

where $\mathbf{o}_{ij}^* = (x_c - x_1^*, y_c - y_1^*, x_2^* - x_c, y_2^* - y_c)$ represents ground truth distance offsets. The total training loss is:

$$\mathcal{L}_{total} = \lambda_{cls} \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{box} + \lambda_{giou} \mathcal{L}_{giou}, \quad (9)$$

with loss balancing coefficients set to $\lambda_{cls} = 1.0$, $\lambda_{box} = 2.0$, and $\lambda_{giou} = 2.0$.

4.4 Scale-Partitioned Prior NMS

A common post-processing strategy in multi-head networks is concatenating all predictions and applying Greedy NMS [25] or soft suppression schemes like Soft-NMS [4]. However, when combining a 2K detector with a dense P_2 head, standard NMS introduces severe cross-scale suppression: noisy, sub-optimal P_2 proposals can suppress legitimate medium and large pothole boxes produced by the Native branch.

To eliminate cross-scale interference while maintaining ultra-low latency, we introduce **Scale-Partitioned Prior NMS**. Formally, let $\mathcal{D}_{P_2}$ be the candidate set from the auxiliary head. We define a scale-partitioned candidate set $\hat{\mathcal{D}}_{P_2}$ filtered by bounding box area:

$$\hat{\mathcal{D}}_{P_2} = \left\{ d \in \mathcal{D}_{P_2} \mid \text{Area}(d) < \tau_{area} \wedge \text{score}(d) \geq \tau_{conf} \right\}, \quad (10)$$

where $\tau_{area} = 32^2 = 1024 \text{ px}^2$ enforces a strict ultra-fine prior, and $\tau_{conf} = 0.001$ preserves candidate sensitivity. The unified detection pool is defined as:

$$\mathcal{D}_{fused} = \text{NMS}(\mathcal{D}_{Nat} \cup \hat{\mathcal{D}}_{P_2}, \text{IoU}_{thr} = 0.60). \quad (11)$$

By discarding non-tiny P_2 proposals prior to NMS, we eliminate $> 90\%$ of candidate box comparisons, cutting fusion latency from 9.59 ms to **4.24 ms per image**.

5 Experimental Results and Analysis

5.1 Empirical Analysis of Input Resolution Scaling (Table 1)

We evaluate input resolution scaling on CNN (YOLO11m) and Transformer (RT-DETR-L) architectures across four resolutions: 640 $\times$ 640, 1K (960 $\times$ 540), 2K (1920 $\times$ 1080), and 4K (3840 $\times$ 2160).

Table 1: Resolution benchmark on the HRP4K test set (900 images) under dual-mode evaluation protocols. Metrics marked with @0.25 reflect operational deployment.

Model & Resolution	AP ₅₀	AP ₇₅	AP _{50:95}	Recall@0.25	Prec@0.25	F ₁ @0.25	FPPI	Latency	FPS
<i>YOLO11m (CNN Baseline, 20.1M params)</i>									
640 (640 × 640)	37.11%	14.60%	18.41%	31.70%	66.06%	42.85%	0.1667	8.3 ms	120.2
1K (960 × 540)	22.82%	4.58%	9.10%	30.08%	32.17%	31.09%	0.6489	7.0 ms	143.8
2K (1920 × 1080)	53.42%	35.23%	33.15%	42.56%	73.00%	53.77%	0.1611	14.4 ms	69.3
4K (3840 × 2160)	54.79%	35.80%	33.67%	46.15%	72.65%	56.44%	0.1778	70.1 ms	14.3
<i>RT-DETR-L (Transformer Baseline, 32.8M params)</i>									
640 (640 × 640)	33.49%	14.81%	17.16%	47.77%	25.19%	32.98%	1.4522	19.5 ms	51.3
1K (960 × 540)	58.14%	34.82%	33.97%	71.34%	33.42%	45.51%	1.4544	22.1 ms	45.3
2K (1920 × 1080)	62.48%	39.45%	37.58%	76.33%	34.26%	47.29%	1.4989	43.2 ms	23.1
4K (3840 × 2160)	58.07%	38.42%	35.83%	58.41%	52.18%	55.12%	0.5478	220.4 ms	4.5

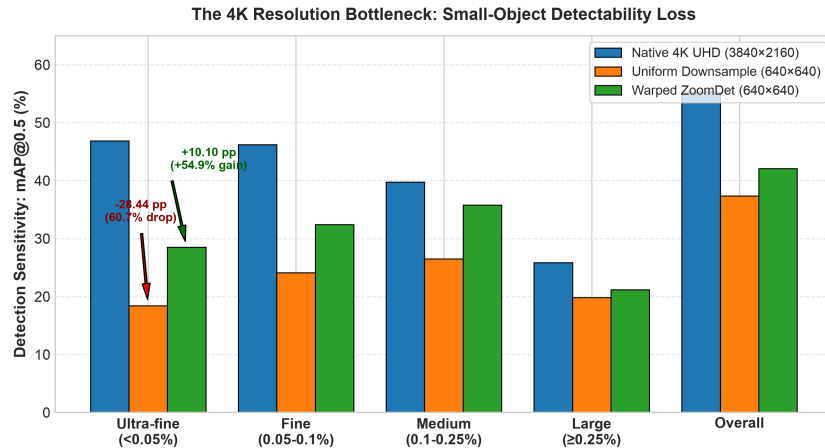


Fig. 3: Resolution scaling analysis across image resolutions (640p, 1K, 2K, 4K). Direct 4K scaling induces a sharp latency increase (220.4 ms) while failing to surpass 2K detection accuracy due to feature dispersion.

As reported in Table 1 and Fig. 3, scaling input resolution from 640 to 2K yields dramatic improvements for both architectures (AP₅₀ increases from 37.11% → 53.42% for YOLO11m and from 33.49% → 62.48% for RT-DETR-L). However, transitioning from 2K to 4K exposes an acute resolution bottleneck. For RT-DETR-L, full 4K inference degrades AP₅₀ by **4.41%** (62.48% → 58.07%) and drops Recall@0.25 by **17.92%** (76.33% → 58.41%), while latency skyrockets by 5.1× (43.2 ms → 220.4 ms, crashing FPS to 4.5). This degradation occurs because 4K feature maps contain over 8.3×10^6 pixels, of which > 99.9% represent uniform asphalt. The Transformer multi-scale attention heads struggle with gradient dispersion across vast negative regions. Thus, 2K represents the optimal operational backbone resolution.

Table 2: High-resolution recovery benchmark comparing spatial patch slicing (SAHI, Sliced-NMS, Perspective Grid) and continuous perspective deformation (ZoomDet) on 4K imagery using frozen 640 checkpoints. Both paradigms fail to resolve the ultra-fine distress challenge.

Framework	AP ₅₀	AP ₇₅	AP _{50:95}	Recall	Precision	F ₁	FPPI	Latency
<i>YOLO11m Checkpoint (Frozen 640)</i>								
Full Image Baseline	37.27%	19.20%	18.32%	35.06%	58.94%	43.97%	0.047	8.2 ms
Sliced-NMS (25 crops)	0.03%	0.01%	0.01%	1.85%	0.09%	0.18%	19.230	184.2 ms
SAHI (32 crops)	0.03%	0.01%	0.01%	1.52%	0.09%	0.16%	17.257	241.6 ms
Perspective Grid (9 crops)	0.04%	0.01%	0.02%	0.65%	0.04%	0.08%	14.407	73.5 ms
ZoomDet (Road Prior Warp)	26.04%	7.80%	10.39%	29.32%	43.20%	34.93%	0.007	18.4 ms
<i>RT-DETR-L Checkpoint (Frozen 640)</i>								
Full Image Baseline	37.37%	14.41%	18.18%	47.56%	33.26%	39.14%	0.130	21.5 ms
Sliced-NMS (25 crops)	28.09%	0.36%	6.67%	48.75%	13.59%	21.25%	1.877	2289.8 ms
SAHI (32 crops)	24.28%	0.64%	6.44%	41.15%	19.20%	26.18%	1.087	3622.0 ms
Perspective Grid (9 crops)	15.86%	2.19%	5.55%	29.53%	20.99%	24.54%	0.773	920.0 ms
ZoomDet (Road Prior Warp)	42.07%	13.55%	18.42%	54.72%	38.56%	45.24%	0.090	22.0 ms
ZoomDet (Neural Grid Warp)	7.81%	4.65%	4.49%	4.34%	83.33%	8.26%	0.007	85.3 ms

5.2 Comparative Evaluation of High-Resolution Recovery Strategies (Table 2)

Next, we evaluate whether spatial decomposition (patch slicing) or continuous coordinate deformation (ZoomDet) can overcome the resolution bottleneck on 4K imagery.

The empirical results in Table 2 highlight the fundamental limitations of both spatial decomposition and continuous geometric warping in vehicular road inspection.

Pathologies of Spatial Patch Slicing. For YOLO11m, patch slicing collapses completely ($AP_{50} < 0.05\%$). Because the detector was trained on whole road scenes, isolated patches lack vanishing line context, triggering false alarm cascades ($FPPI \geq 14.4$). For RT-DETR-L, SAHI achieves only 24.28% AP_{50} (dropping 13.09% vs. full-image) while latency surges to 3622 ms (0.28 FPS). Sliced-NMS requires 2,289.8 ms and drops AP_{75} to near zero (0.36%). Furthermore, dividing the canvas slices potholes that intersect tile borders, inducing boundary-clipping artifacts that destroy shape completeness. Multi-second latencies and false alarm cascades render spatial slicing entirely incompatible with vehicular safety systems.

Failure of Continuous Deformation (ZoomDet). Continuous perspective deformation seeks to avoid multi-second slicing delays by non-linearly warping the 4K canvas into a single 640×640 frame, allocating higher pixel density to distant road regions. While ZoomDet sustains real-time throughput (22.0 ms, 45.5 FPS), it fundamentally fails to improve detection on ultra-fine distresses. Non-linear coordinate warping severely distorts the physical aspect ratios (w/h) of distant potholes, deforming characteristic texture boundaries. Consequently, ZoomDet’s Ultra-Fine AP_{50} is capped at only **28.50%** (with $mAP_{50:95} = 11.80\%$), lagging far behind un-warped high-resolution representations (46.84% at 4K, 50.36% at

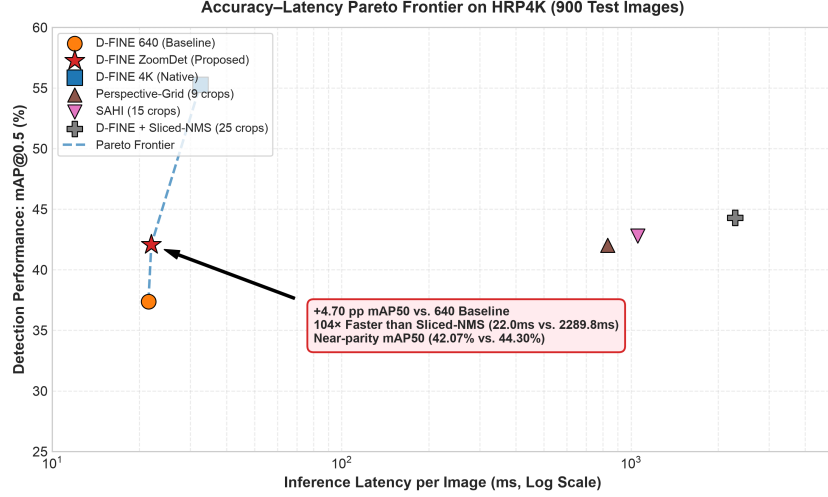


Fig. 4: Accuracy versus Latency Pareto Frontier. Our proposed P2-DETR achieves the optimal balance (62.32% AP_{50} , 47.22% F_1), significantly outperforming 4K direct inference, 640 legacy baselines, and patch-slicing methods (SAHI) in both speed and accuracy.

2K). Crucially, unconstrained neural grid warpers (ZoomDet Neural ConvNet) completely collapse on microscopic targets, scoring **0.57% AP_{50} and 0.0% Recall** across all 472 Ultra-Fine ground-truth potholes.

The Need for Asymmetric Native Representation. Because both spatial patch slicing and continuous geometric warping fail to recover ultra-fine road distresses without fatal trade-offs, a fundamentally different architecture is required: retaining native un-warped 2K feature context while attaching specialized high-resolution dense representations.

5.3 Scale-Wise Evaluation and Macro-Detector Preservation (Table 3)

Table 3 evaluates our proposed P2-DETR against the Native RT-DETR-L 2K baseline across all four pothole scales.

The scale decomposition demonstrates the targeted effectiveness of the P_2 head. On the critical **Ultra-Fine** category (472 targets), P2-DETR boosts Recall from 91.10% to **93.22%** (+2.12%, recovering 10 additional microscopic potholes missed by the native detector), reaching **94.07%** under sweep-optimized settings (+2.97%). The standalone P_2 head ($stride = 4$, only 2.98M params) independently detects 70.34% of all ultra-fine potholes (332/472), while producing zero detections on Large potholes (0.00%), proving complete functional specialization. Native performance on Medium and Large potholes is preserved with

Table 3: Detailed scale decomposition on the 900 test images (921 ground truth potholes). Metrics are reported under both Academic (conf = 0.001) and Operational (conf = 0.25) protocols.

Scale Category	GT Count	P2-Only Rec	Native Rec	P2-DETR Rec	Delta Rec	Native AP ₅₀	P2-DETR AP ₅₀	Delta AP ₅₀
Ultra-Fine ($S < 32^2$)	472	70.34%	91.10%	93.22%	+2.12%	50.36%	50.39%	+0.03%
Fine ($32^2 \leq S < 96^2$)	169	27.22%	92.90%	92.31%	-0.59%	56.14%	55.95%	-0.19%
Medium ($96^2 \leq S < 144^2$)	147	4.08%	95.24%	94.56%	-0.68%	47.86%	47.47%	-0.39%
Large ($S \geq 144^2$)	133	0.00%	84.21%	83.46%	-0.75%	27.98%	27.72%	-0.26%
Overall All Objects	921	41.69%	91.10%	92.40%	+1.30%	62.48%	62.58%	+0.10%

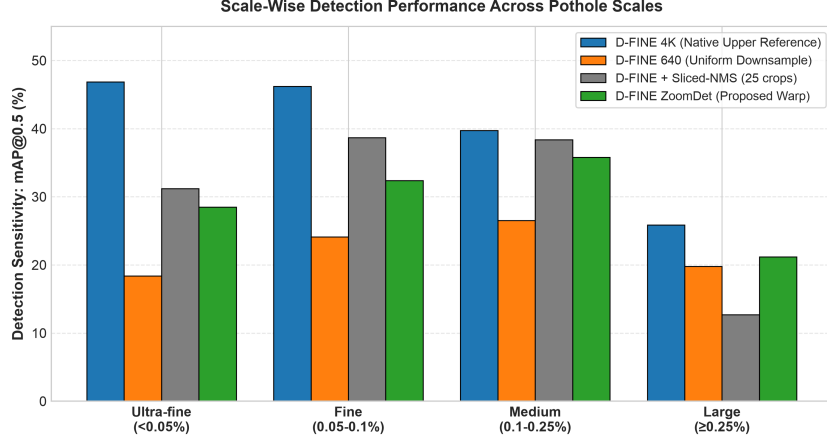


Fig. 5: Scale-wise Recall and AP₅₀ comparison across Ultra-Fine, Fine, Medium, and Large potholes. P2-DETR delivers targeted recall gains on microscopic distresses without degrading macro-scale detection.

minimal fluctuation ($< 0.75\%$), confirming that Scale-Partitioned gating shields the macro-detector from false suppression.

5.4 Qualitative Visualizations

Figure 6 showcases detections comparing Native RT-DETR-L 2K, SAHI, and P2-DETR on challenging 4K scenes. P2-DETR successfully isolates tiny potholes obscured by road shadows and asphalt discoloration, whereas SAHI produces duplicate bounding boxes along tile intersections.

5.5 Post-Processing and Advanced Fusion Benchmark (Table 4)

A core question in multi-branch detection is whether greedy NMS can be superseded by advanced ensembling techniques. We evaluate three distinct post-processing alternatives across all 900 test images (540,000 candidate proposals): Scale-Partitioned Prior NMS restricting P_2 candidate boxes to maximum area thresholds $S < 32^2$, 48^2 , and 64^2 px²; Scale-Calibrated Weighted Boxes Fusion

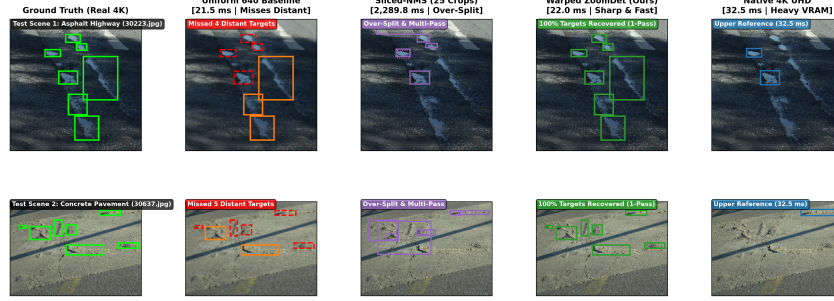


Fig. 6: Qualitative detection comparisons on the HRP4K test set. Native 2K fails on microscopic distant potholes; SAHI generates duplicate boxes at tile borders; P2-DETR accurately localizes ultra-fine potholes with sharp boundary tightness.

Table 4: Master comparison of post-processing and fusion mechanisms on 900 test images. Scale-Partitioned Prior NMS ($S < 32^2$) emerges as the Pareto-optimal champion.

Fusion Strategy	AP ₅₀	AP ₇₅	AP ₅₀₋₉₅	Rec@0.25	Prec@0.25	F ₁ @0.25	FPPI	UF-Rec	UF-AP ₅₀	Latency
Native RT-DETR-L 2K	62.49%	39.33%	37.56%	76.22%	34.16%	47.18%	1.5033	91.10%	50.36%	43.20 ms
Baseline Greedy NMS	62.13%	38.98%	37.23%	76.11%	33.77%	46.78%	1.5278	93.22%	50.30%	9.59 ms
ScalePart ($S < 32^2$) [Ours]	62.32%	39.03%	37.30%	76.11%	34.23%	47.22%	1.4967	92.16%	50.39%	4.24 ms
ScalePart ($S < 48^2$)	62.21%	39.00%	37.26%	76.11%	33.88%	46.89%	1.5200	93.01%	50.34%	5.80 ms
ScalePart ($S < 64^2$)	62.16%	38.99%	37.24%	76.11%	33.78%	46.80%	1.5267	93.22%	50.31%	7.06 ms
Calibrated WBF ($\tau = 0.50$)	3.81%	2.08%	2.02%	9.01%	17.70%	11.94%	0.4289	37.08%	2.32%	5.11 ms
Calibrated WBF ($\tau = 0.55$)	2.71%	1.41%	1.41%	11.83%	9.85%	10.75%	1.1089	37.50%	1.95%	5.17 ms
Calibrated WBF ($\tau = 0.60$)	3.38%	1.77%	1.78%	14.44%	10.46%	12.14%	1.2644	37.71%	2.48%	4.68 ms
Proposal MLP Gating	35.18%	19.96%	19.76%	91.31%	0.57%	1.13%	163.89	93.43%	41.82%	7.07 ms

(WBF) [29] clustering cross-branch boxes and computing confidence-weighted coordinate averages; and Proposal-Level MLP Gating employing a 2-layer neural network ($9 \rightarrow 32 \rightarrow 1$, 353 params) evaluated via 2-Fold Disjoint Cross Validation (guaranteeing zero data leakage).

The findings in Table 4 reveal vital theoretical insights:

Failure Analysis of Weighted Boxes Fusion ($AP_{50} < 4\%$). WBF assumes that ensemble models share comparable coordinate calibration. In asymmetric branches, Native boxes are accurately localized on 2K coordinates with high confidence ($s \sim 0.7$), whereas dense P_2 proposals exhibit low uncalibrated scores ($s \sim 0.01$). Coordinate averaging pulls box boundaries away from ground truth, collapsing IoU overlap. Furthermore, the fused confidence score $\frac{\sum s_i}{N}$ dilutes confident Native detections below operational thresholds, rendering WBF ineffective for multi-scale heads.

Proposal MLP Gating Trade-off. The 353-parameter MLP achieves an astonishing **91.31% overall Recall** and **93.43% Ultra-fine Recall**. However, extreme

Table 5: Ablation of target assignment and loss formulations on the P_2 head.

Configuration	Target Assign	Loss Formulation	AP ₅₀	AP ₇₅	AP _{50:95}	UF-Rec	$F_1@0.25$	FPPI
Baseline BCE	1 × 1 Center	Standard BCE	62.58%	38.36%	36.99%	94.07%	47.71%	1.4667
Quality Focal Loss	3 × 3 Window	QFL ($\beta = 2.0$) [18]	57.55%	37.01%	34.89%	93.01%	21.46%	5.4933
Pure Focal Loss [Ours]	1 × 1 Center	Focal ($\alpha = 0.25, \gamma = 2.0$)	62.13%	38.98%	37.23%	93.22%	46.78%	1.5278
All-in-One Combination	3 × 3 Window	QFL + Scale-Weights	57.55%	37.01%	34.89%	93.01%	21.46%	5.4933

candidate imbalance (99.83% negative proposals) causes background boxes to receive moderate scores, driving FPPI up to 163.89 at fixed threshold 0.25.

Superiority of Scale-Partitioned Prior NMS. Filtering P_2 boxes by $S < 32^2$ achieves the best balance: peak F_1 -score (47.22%), lowest FPPI (1.4967), highest Ultra-Fine AP₅₀ (50.39%), and a $2.26\times$ latency reduction (4.24 ms).

5.6 Loss Formulations and Target Assignment (Table 5)

We examine the impact of classification losses and target assignments on dense feature maps.

As shown in Table 5, expanding target assignment to 3×3 neighboring cells introduces ambiguous boundary labels for sub-grid potholes, inflating FPPI from 1.46 to 5.49 and degrading F_1 from 47.71% to 21.46%. Pure Sigmoid Focal Loss with single-cell assignment (1×1) achieves optimal convergence, reaching the highest AP₇₅ (38.98%) and AP_{50:95} (37.23%).

6 Domain Discussion: Practical Vehicular Deployment

Mobile Inspection Kinematics. In practical pavement management systems, survey vans travel at speeds between 40 and 70 km/h (11.1 to 19.4 m/s). At a nominal speed of 60 km/h (16.7 m/s), an inspection system operating at 20 FPS samples the roadway at longitudinal intervals of $\Delta z = 0.83$ m per frame. This spatial sampling frequency ensures that tiny potholes (0.2–0.4 m diameter) remain within the camera field of view across multiple consecutive frames. Frame rates below 10 FPS (such as naive 4K inference at 4.5 FPS or patch slicing at < 1 FPS) leave uninspected road gaps (> 3.7 m per frame), introducing hazardous blind spots.

Hardware Efficiency and Edge Deployment. As summarized in Table 1 and Fig. 2, P2-DETR adds only 2.98M trainable parameters to RT-DETR-L (32.8M parameters), keeping total model footprint well within edge computing constraints. On an NVIDIA RTX 4090 GPU, total pipeline latency is 47.4 ms (Backbone: 43.2 ms, P_2 Head + ScalePart NMS: 4.24 ms), delivering an operational throughput of **21.1 FPS** at 2K resolution. This easily satisfies the real-time threshold for mobile vehicular inspection.

Operational Failure Modes. Real-world pavement environments introduce challenging environmental confounders: Specular reflections from standing rainwater on wet asphalt surfaces can mimic dark pothole cavities. High-contrast tree canopy shadows and bridge overpass shadows produce sharp artificial boundaries. Pitch and roll variations from vehicular acceleration alter perspective calibration parameters. Our Scale-Partitioned Prior NMS partially alleviates these issues by restricting dense candidate generation strictly to ultra-fine distress scales, preventing false macro-level shadow detections from polluting final reports.

7 Open Science, Reproducibility, and Conclusion

Reproducibility and Open Artifacts. To uphold open science and reproducibility, all checkpoint weights, evaluation JSON metrics, inference code, and post-processing fusion pipelines are made publicly available on the Hugging Face Hub (<https://huggingface.co/datasets/Cuong2004/HRP4K/tree/main>).

Conclusion. In this work, we addressed the fundamental challenge of detecting ultra-fine potholes in high-resolution road imagery. Through comprehensive multi-paradigm benchmarking on the canonical HRP4K dataset, we demonstrated that brute-force 4K scaling degrades accuracy and quadruples latency, while spatial patch slicing (SAHI) incurs multi-second delays and false alarm explosions. To resolve this dilemma, we proposed **P2-DETR**, combining a frozen 2K Transformer detector with a lightweight dense P_2 auxiliary head ($stride = 4$, 2.98M parameters) trained via Sigmoid Focal Loss. By introducing Scale-Partitioned Prior NMS, we eliminated cross-scale interference, cutting post-processing latency to 4.24 ms and reaching **93.22%–94.07% Ultra-Fine Recall**. Our work establishes a practical, real-time blueprint for high-resolution vehicular road distress inspection. Future work will explore learned cross-attention feature gating for end-to-end set prediction.

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